

**AMENDMENT NO. 1      JANUARY 1988**  
**TO**  
**IS : 6071 - 1986   SPECIFICATION FOR**  
**SYNTHETIC SEPARATORS FOR LEAD-ACID**  
**BATTERIES**

**( *First Revision* )**

( *Page 4, clause 6.1, line 3* ) — Substitute '0.75 ohms/cm<sup>2</sup> for '0.75 ohms cm<sup>2</sup>/mm of thickness ( web thickness in case of ribbed separators )'.

( *Page 6, clause 14* ) — Substitute the following for the existing clause:

**'14. TEST FOR WETTABILITY**

**14.1** Place a drop of 1.280 ( 27°C ) sulphuric acid solution with a pipette ( 10 cc ) on the surface of the separators at room temperature.

NOTE — Equal size of drops shall always be used. The height from which the drop is placed should be approximately 1.5 cm. This is to ensure correct size of drop so that the liquid does not splash.

**14.2** The drop shall be absorbed by the separators within 60 s.

**14.3** Test shall be carried out on both the surfaces of the separators.'

( *Page 6, clause 16.2.1* ) — Substitute the following, for the existing clause:

**'16.2.1** Each package may also be marked with the Standard Mark.

NOTE — The use of the Standard Mark is governed by the provisions of the Bureau of Indian Standards Act 1986 and the Rules and Regulations made thereunder. The Standard Mark on products covered by an Indian Standard conveys the assurance that they have been produced to comply with the requirements of that standard under a well-defined system of inspection, testing and quality control which is devised and supervised by BIS and operated by the producer. Standard marked products are also continuously checked by BIS for conformity to that standard as a further safeguard. Details of conditions under which a licence for the use of the Standard Mark may be granted to manufacturers or producers, may be obtained from the Bureau of Indian Standards.'

( *Page 6, clause 16.2.1* ) — Add the following new clauses after **16.2.1**:

**17. TESTS AND PERFORMANCE**

**17.1 Classification of Tests**

**17.1.1 Type Tests** — The following shall constitute the type tests:

- a) Electrical resistance ( **6.1** ),

- b) Chlorine content ( 7.1 ),
- e) Iron content ( 8.1 ),
- d) Manganese content ( 9.1 ),
- e) Maximum predominant pore size ( 10.1 and 10.2 ),
- f) Volume porosity ( 11.1 and 11.2 ),
- g) Life test ( 12.1 and 12.2 ),
- h) Test for mechanical test ( 13 ), and
- j) Test for wettability of separator ( 14.1, 14.2 and 14.3 ).

**17.1.2 Acceptance Tests** — The following tests shall constitute as acceptance tests:

- a) Dimensions ( 5.1 ),
- b) Wettability of separator ( 14.1 to 14.3 ), and
- c) Volume porosity ( 11.1 and 11.2 ).

**17.1.2.1** The acceptance tests shall be applied on two samples up to a maximum of 1 percent of each type in a lot, the samples being drawn at random by the purchasing or the inspecting authority,

**17.1.2.2 Criteria for acceptance** — If any of the sample separator fails in any of the acceptance tests, twice the original number of samples shall be taken and subjected to all the acceptance tests. If there is any failure in the retests, the lot may be rejected.'

( Page 8, clause **A-1.2.4** ) — Substitute the following for the existing clause:

**'A-1.2.4** Report the difference in resistance with and without the hole in separator, as ohms square centimetre of the surface area of the separator. The resistance of the separator is then given by the following formula:

$$( O_w - O_H ) \times A \text{ ohms' cm}^3$$

where

$O_w$  = resistance in ohms of separator without hole,

$O_H$  = resistance in ohms of separator with hole, and

$A$  = area of the hole in  $\text{cm}^2$ .'

( Page 10, clause **A-2.2.1**, last sentence ) — Substitute the following for the existing sentence:

'The resistance is measured 2 hours after immersion.'

( Page 11, clause **A-2.2.4** ) — Substitute the following for the existing clause:

**'A-2.2.4** The resistance is expressed in terms of  $\text{ohm.cm}^2$  of the area of the separator.'

( Page 11, clouse **A-2.2.4** ) — Add the following new clause after **A-2.2.4**:

'**A-3.** Alternatively any other standard equipment based on the same principle in accordance with **A-1** and **A.2** can also be used.'

( Page 12, clause **B-1.2** ) — Delete and renumber the subsequent clauses accordingly.

( Pape 12, clause **B-2.2** ) — Substitute the following for the existing clause:

'**B-2.2** Transfer 100 ml of the aliquot ( see **B-2.1** ) into a 600-ml conical flask. Cool, add exactly 10.0 ml of standard silver nitrate solution. Add a few millilitres of nitrobenzene and shake to coagulate the precipitate of silver chloride. Titrate the excess of silver nitrate with the standard ammonium thiocyanate solution using ferric ammonium sulphate solution as indicator. The end point of the titration is a faint permanent brown colouration which is difficult to see without considerable experience. If any doubt is felt about the end-point, it should be compared against a similar solution containing dilute sulphuric acid, nitrobenzene, ferric ammonium sulphate solution and one drop of standard ammonium thiocyanate solution which gives the colour of the end-point.'

( ETDC 11 )